

Internship Assignment:

Task 1

Business Intelligence Case Study

Objective

You have recently joined the Business Intelligence team as a Data Analyst Intern. Your task is to analyze a raw business dataset, identify key business problems, create meaningful KPIs, design an interactive dashboard, and present actionable insights to management.

Unlike a typical dashboard assignment, **you will not be told what charts or KPIs to create**. Your responsibility is to understand the business and decide what information will help management make better decisions.

Business Scenario

TechNova Retail Pvt. Ltd. is a fast-growing retail company operating across India. The company sells products through both online and offline channels.

Recently, the management observed that although revenue has been increasing, profitability is declining in certain regions. They are unsure about:

- Which products generate the highest profits?
- Which products are causing losses?
- Which customers contribute the most revenue?
- Are discounts affecting profitability?
- Which cities or states perform well?
- Which sales executives need improvement?
- Which categories should receive more investment?
- How does seasonality affect sales?

The management has asked you to prepare a dashboard to answer these business questions.

Dataset Link :

<https://docs.google.com/document/d/1PgZ6AGXX71V28qMWl9nChW5uoMxKr6iyvuSZNsAZeZs/edit?usp=sharing>

Dashboard Requirements

The dashboard should contain:

- Minimum 12 KPIs
- Minimum 12 visuals
- Interactive filters
- Drill-through page
- Tooltips
- Navigation buttons
- Consistent theme
- Responsive layout
- Executive Summary page
- Detailed Analysis page

Task 2

Workflow Automation using CrewAI / AutoGen

Objective

Design and implement an AI-powered workflow automation system using **CrewAI** or **Microsoft AutoGen** that simulates a real-world business process involving multiple AI agents.

The system should demonstrate how multiple agents collaborate to complete a task with minimal human intervention.

Business Scenario

A company receives hundreds of internship applications every week. The HR team spends significant time reviewing resumes, preparing interview questions, and sending candidate updates.

Your task is to automate this workflow using AI agents.

Step	AI Agent	Responsibility	Output
1	Resume Analyzer Agent	Reads and analyzes the candidate's resume to extract personal, educational, and professional information.	Resume Summary
2	Skill Extraction Agent	Identifies technical skills, soft skills, tools, programming languages, frameworks, and certifications.	Skills Profile
3	Candidate Evaluation Agent	Evaluates the candidate based on predefined criteria and assigns a suitability score.	Candidate Score & Evaluation
4	Interview Question Generator	Generates role-specific HR, technical, coding, and scenario-based interview questions.	Interview Question Set

5	HR Decision Agent	Reviews the evaluation and recommends whether the candidate should be shortlisted, kept on hold, or rejected.	Hiring Recommendation
6	Email Draft Generator	Creates an appropriate email based on the hiring decision (Interview Invitation, Shortlist, or Rejection).	Email Draft
7	Report Generator	Compiles all outputs into a structured report for HR review.	Final Candidate Assessment Report